

TP3 • COMMUNICATION PROTOCOLS

Lab 4 — 100GBASE-R PCS & Multi-Lane Distribution (MLD)

Vasquez Curetti, Francisco Javier

IEEE 802.3 Clause 82: how real 100GBASE-R MLD works

01 Round-robin distribution. The 66b block stream is spread across 20 Virtual Lanes (VLs) in sequence, to run over multiple physical channels (e.g. 4×25G, 10×10G).

02 Alignment Markers (AM). Every 16383 blocks per lane, an AM replaces a real data block — a unique 66-bit pattern per lane, carrying a BIP3/BIP7 parity check.

03 Deskew & reorder. The receiver buffers each lane until every AM lines up, then reconstructs the single stream using the lane ID embedded in the marker.

Table 82-2 — Alignment Marker encoding per lane

Table 82-2—100GBASE-R Alignment marker encodings

PCS lane number	Encoding ^a [M ₀ , M ₁ , M ₂ , BIP ₃ , M ₄ , M ₅ , M ₆ , BIP ₇]	PCS lane number	Encoding ^a [M ₀ , M ₁ , M ₂ , BIP ₃ , M ₄ , M ₅ , M ₆ , BIP ₇]
0	0xC1, 0x68, 0x21, BIP ₃ , 0x3E, 0x97, 0xDE, BIP ₇	10	0xFD, 0x6C, 0x99, BIP ₃ , 0x02, 0x93, 0x66, BIP ₇
1	0x9D, 0x71, 0x8E, BIP ₃ , 0x62, 0x8E, 0x71, BIP ₇	11	0xB9, 0x91, 0x55, BIP ₃ , 0x46, 0x6E, 0xAA, BIP ₇
2	0x59, 0x4B, 0xE8, BIP ₃ , 0xA6, 0xB4, 0x17, BIP ₇	12	0x5C, 0x B9, 0xB2, BIP ₃ , 0xA3, 0x46, 0x4D, BIP ₇
3	0x4D, 0x95, 0x7B, BIP ₃ , 0xB2, 0x6A, 0x84, BIP ₇	13	0x1A, 0xF8, 0xBD, BIP ₃ , 0xE5, 0x07, 0x42, BIP ₇
4	0xF5, 0x07, 0x09, BIP ₃ , 0x0A, 0xF8, 0xF6, BIP ₇	14	0x83, 0xC7, 0xCA, BIP ₃ , 0x7C, 0x38, 0x35, BIP ₇
5	0xDD, 0x14, 0xC2, BIP ₃ , 0x22, 0xEB, 0x3D, BIP ₇	15	0x35, 0x36, 0xCD, BIP ₃ , 0xCA, 0xC9, 0x32, BIP ₇
6	0x9A, 0x4A, 0x26, BIP ₃ , 0x65, 0xB5, 0xD9, BIP ₇	16	0xC4, 0x31, 0x4C, BIP ₃ , 0x3B, 0xCE, 0xB3, BIP ₇
7	0x7B, 0x45, 0x66, BIP ₃ , 0x84, 0xBA, 0x99, BIP ₇	17	0xAD, 0xD6, 0xB7, BIP ₃ , 0x52, 0x29, 0x48, BIP ₇
8	0xA0, 0x24, 0x76, BIP ₃ , 0x5F, 0xDB, 0x89, BIP ₇	18	0x5F, 0x66, 0x2A, BIP ₃ , 0xA0, 0x99, 0xD5, BIP ₇
9	0x68, 0xC9, 0xFB, BIP ₃ , 0x97, 0x36, 0x04, BIP ₇	19	0xC0, 0xF0, 0xE5, BIP ₃ , 0x3F, 0x0F, 0x1A, BIP ₇

^aEach octet is transmitted LSB to MSB.

DESIGN

M4=~M0, M5=~M1, M6=~M2 for every lane (e.g. lane 0: 0xC1 ↔ 0x3E, 0x68 ↔ 0x97, 0x21 ↔ 0xDE) — each marker carries its own bitwise complement as a built-in check, on top of the BIP3/BIP7 fields.

This lab’s abstraction vs. the real Clause 82 mechanism

Aspect	Real spec (Clause 82)	This lab's RTL
AM interval	every 16383 blocks	every 64 blocks (AM_INTERVAL param)
Lane data path	20 physical FIFOs, real demux + skew	1 continuous stream, blocks never split
AM insertion	replaces a real block; needs idle-rate compensation	never replaces pcs_block_out — rides alongside as sideband wires
Lane ID	implicit; decoded from the 8-byte marker itself	explicit byte inside a fixed template (0xC1+laneID+pad)
BIP3/BIP7	real running parity, detects bit errors	not implemented — fixed 0x00/0xA5 filler

Test setup: dual-namespace TAP + traffic generation

01

Two Linux network namespaces. tap0 lives in the host namespace (10.0.0.1/24); tap1 lives inside ns_b (10.0.0.2/24). A single emulator process binds both ends via setns().

02

Build parameters. -GAM_INTERVAL=64 -GNUM_LANES=20. FST tracing is unconditional here — dump.fst is produced on every run, no build flag needed.

03

Traffic. ping -c 5 -p fec80004 -I tap0 10.0.0.2 — ARP + 5 ICMP echo pairs.

End-to-end integrity

▶ Frame 3: 98 bytes on wire (784 bits), 98 bytes captured (784 bits)

▼ Ethernet II, Src: 02:ca:ee:44:4a:fa (02:ca:ee:44:4a:fa), Dst: b6:cb:dc:76:cc:85 (b6:cb:dc:76:cc:85)

▶ Destination: b6:cb:dc:76:cc:85 (b6:cb:dc:76:cc:85)

▶ Source: 02:ca:ee:44:4a:fa (02:ca:ee:44:4a:fa)

Type: IPv4 (0x0800)

▶ Internet Protocol Version 4, Src: 10.0.0.1, Dst: 10.0.0.2

▶ Internet Control Message Protocol

0000b6cbdc76cc8502caee444afa08004500...v...DJ...

00100054733a40004001b36c0a0000010a00Ts:@@l...

002000020800061e1d68000119f29e6a0000.....h...

003000002006080000000000fe ca0004fe ca.....

00400004fe ca0004fe ca0004fe ca0004fe ca.....

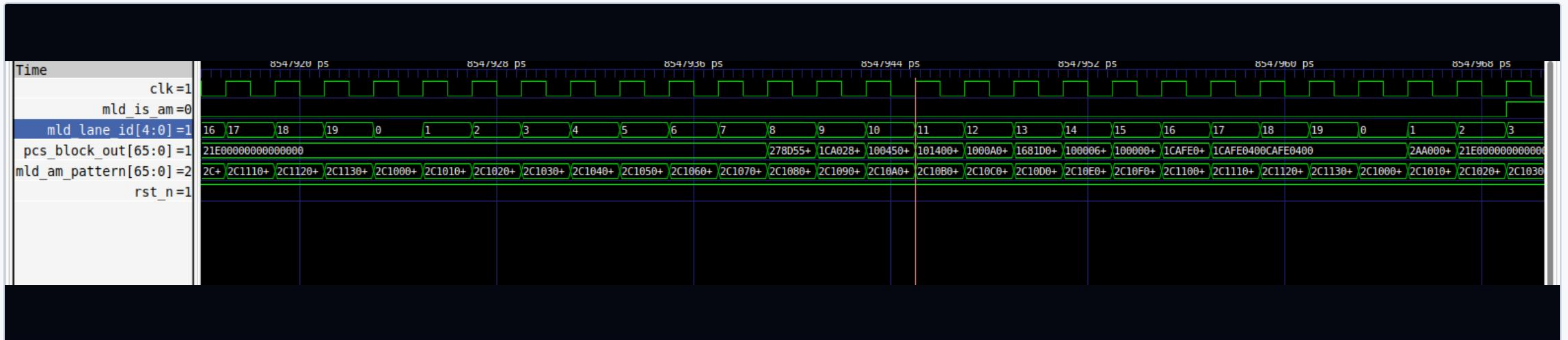
00500004fe ca0004fe ca0004fe ca0004fe ca.....

00600004.....

RESULT

ICMP echo request/reply between 10.0.0.1 and 10.0.0.2, feca0004 payload pattern

Virtual Lane round-robin: mld_lane_id cycles 0 → 19 every clock



ANSWER Q1.1/Q1.2

One 66-bit block dispatched per clock edge; lane_ptr wraps 19 → 0 with NUM_LANES=20 — sequential increment, one block per lane per cycle, no stalls.

Alignment Marker periodicity: 128 ps = 64 cycles between pulses

$t \approx 129 \text{ ps} - \text{pulse 1} \cdot \text{mld_lane_id} = 3$



$t \approx 257 \text{ ps} - \text{pulse 2} \cdot \text{mld_lane_id} = 7 \cdot \Delta t = 128 \text{ ps} = 64 \text{ cycles}$



Why VL0 and VL5 never appear at an AM pulse (Q2.2)

01

Phase lock. AM_INTERVAL=64 and NUM_LANES=20 share gcd=4. From the reset state, mld_is_am only ever coincides with $\text{lane_ptr} \equiv 3 \pmod{4}$.

02

Verified in the trace. Pulses land on lanes {3,7,11,15,19} only, repeating every 5 pulses (320 cycles). Lanes 0 and 5 ($\equiv 0/1 \pmod{4}$) are structurally unreachable at an AM instant.

Real ping payload flowing untouched through the pipeline

